

AARAV SHARMA

Machine Learning Engineer | AI Engineer | Generative AI | MLOps

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PROFESSIONAL SUMMARY

Machine Learning Engineer with broad experience across machine learning, deep learning, NLP, computer vision, generative AI, large language models, recommendation systems, and MLOps. Experienced in designing end-to-end ML systems from data ingestion and feature engineering through model development, evaluation, deployment, monitoring, and optimization. Proficient in Python, SQL, PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, LLM APIs, vector databases, MLflow, Docker, Kubernetes, AWS, GCP, and CI/CD.

TECHNICAL SKILLS

Programming:	Python, Java, C++, JavaScript, TypeScript, SQL, Bash, R, OOP, DSA, Functional Programming, Multithreading, Concurrency
Machine Learning:	Supervised/Unsupervised/Semi-Supervised Learning, Reinforcement Learning, Classification, Regression, Clustering, Feature Engineering, Feature Selection, Ensemble Learning, Transfer Learning, Anomaly Detection, Time Series, Recommendation, Ranking, Hyperparameter Optimization, Cross-Validation, Imbalanced Learning, Explainable AI
Algorithms:	Linear/Logistic Regression, Ridge, Lasso, Elastic Net, Decision Trees, Random Forest, Gradient Boosting, XGBoost, LightGBM, CatBoost, SVM, KNN, Naive Bayes, K-Means, DBSCAN, PCA, t-SNE, UMAP, Isolation Forest, GMM
Deep Learning:	PyTorch, TensorFlow, Keras, CNNs, RNNs, LSTMs, GRUs, Autoencoders, VAEs, GANs, Transformers, Attention, ViTs, Siamese Networks, Contrastive Learning, Self-Supervised Learning, Distillation, Quantization, Pruning, Fine-Tuning
NLP:	Text Classification, Sentiment Analysis, NER, Tokenization, POS Tagging, Summarization, QA, Translation, Topic Modeling, Information Extraction, Semantic Search, TF-IDF, Word2Vec, GloVe, FastText, BERT, RoBERTa, DistilBERT, T5, GPT
GenAI / LLM:	LLMs, Generative AI, Prompt Engineering, Few/Zero-Shot Learning, Structured Outputs, Function Calling, Tool Use, RAG, AI Agents, LLM Evaluation, Fine-Tuning, PEFT, LoRA, QLoRA, RLHF, DPO, Synthetic Data, Guardrails, Hallucination Detection, Observability
LLM Ecosystem:	Hugging Face Transformers/Datasets, LangChain, LangGraph, LlamaIndex, OpenAI APIs, Anthropic APIs, Gemini APIs, Ollama, vLLM, Sentence Transformers, PEFT, TRL, Accelerate
Vector Search:	FAISS, Pinecone, Weaviate, ChromaDB, Milvus, Elasticsearch, OpenSearch, pgvector, ANN Search, Hybrid Search, Semantic Search, Reranking
Computer Vision:	OpenCV, OCR, Image Classification, Object Detection, Segmentation, Image Embeddings, Image Similarity, Image Generation, YOLO, Faster R-CNN, Mask R-CNN, ResNet, EfficientNet, MobileNet, Vision Transformers
Data Engineering:	ETL/ELT, Data Cleaning, Validation, Transformation, Warehousing, Batch/Stream Processing, Apache Spark, PySpark, Kafka, Airflow, dbt, Databricks
Databases:	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, SQLite, Cassandra, DynamoDB, Indexing, Query Optimization, Data Modeling
MLOps:	Experiment Tracking, Model Registry, Versioning, Deployment, Monitoring, Retraining, Drift Detection, Feature Stores, MLflow, Kubeflow, DVC, Weights & Biases, Evidently AI
Cloud:	AWS EC2, S3, Lambda, ECS, EKS, SageMaker, RDS, DynamoDB, CloudWatch, IAM, API Gateway; GCP Vertex AI, Compute Engine, Cloud Storage, BigQuery, Cloud Run, GKE, Pub/Sub; Azure ML, Functions, Blob Storage, AKS, Azure OpenAI
DevOps:	Docker, Kubernetes, Helm, Terraform, GitHub Actions, Jenkins, GitLab CI/CD, REST, GraphQL, FastAPI, Flask, gRPC, Nginx
Software Engineering:	Git, GitHub, Unit Testing, Integration Testing, PyTest, TDD, Code Review, Clean Code, SOLID, Design Patterns, RESTful Architecture, API Design, System Design, Agile/Scrum
Math / Stats:	Linear Algebra, Probability, Statistics, Calculus, Optimization, Gradient Descent, Bayesian Methods, Hypothesis Testing, Confidence Intervals, A/B Testing, MLE, Regularization

PROFESSIONAL EXPERIENCE

Machine Learning Engineer — AI Technologies Pvt. Ltd., Bengaluru | Jan 2024 – Present

- Designed and deployed end-to-end ML pipelines covering preprocessing, feature engineering, training, validation, deployment, and monitoring.
- Developed classification, regression, ranking, recommendation, and anomaly-detection models using Scikit-learn, XGBoost, LightGBM, and PyTorch.
- Built production NLP systems for text classification, semantic search, entity extraction, document understanding, and question answering.
- Developed LLM applications using RAG, embeddings, vector databases, prompt engineering, function calling, and tool-using agents.
- Built RAG pipelines with Hugging Face, LangChain, FAISS, and PostgreSQL/pgvector; implemented retrieval, reranking, query rewriting, and evaluation.
- Fine-tuned transformer models using LoRA and QLoRA for domain-specific NLP tasks.
- Built production inference APIs using FastAPI and Docker and implemented ML experiment tracking with MLflow.
- Developed automated training/deployment pipelines with GitHub Actions and cloud infrastructure; monitored latency, drift, prediction quality, and resources.
- Built data pipelines using Python, SQL, Airflow, Spark, and Kafka; optimized inference through batching, caching, quantization, and efficient preprocessing.
- Worked with AWS services including S3, EC2, SageMaker, Lambda, ECS, EKS, RDS, and CloudWatch.

SELECTED PROJECTS

Enterprise RAG & AI Knowledge Assistant

Python, PyTorch, Hugging Face, LangChain, FAISS, PostgreSQL, FastAPI, Docker, AWS

- Built document ingestion, chunking, metadata extraction, embedding, vector indexing, retrieval, reranking, and response generation.
- Implemented semantic/hybrid search, conversational memory, query rewriting, citation-aware responses, and evaluation for retrieval quality, faithfulness, and hallucination.

Real-Time Fraud Detection System

Python, XGBoost, Kafka, Spark, Redis, PostgreSQL, MLflow

- Built streaming fraud detection with behavioral/transactional features, Kafka event processing, Redis low-latency features, model monitoring, drift detection, and automated retraining.

Computer Vision Object Detection Platform

Python, PyTorch, OpenCV, YOLO, Docker, Kubernetes

- Developed real-time image/video detection, augmentation, YOLO fine-tuning, batching, quantization, and scalable Kubernetes inference services.

Personalized Recommendation Engine

Python, Scikit-learn, PyTorch, PostgreSQL, Redis

- Built hybrid collaborative/content-based recommendations, user/item embeddings, candidate generation, ranking, caching, and evaluation using Precision@K, Recall@K, MAP, NDCG, and CTR.

LLM-Powered AI Agent

Python, LangGraph, OpenAI API, FastAPI, PostgreSQL

- Built a tool-using agent with planning, tool selection, structured outputs, function calling, state management, context retrieval, prompt-injection safeguards, tracing, and evaluation.

EDUCATION

Bachelor of Engineering — Computer Science | University Name, India | 2020 – 2024

Coursework: Machine Learning, Artificial Intelligence, Deep Learning, DSA, DBMS, Operating Systems, Computer Networks, Distributed Systems, Cloud Computing, Statistics, Linear Algebra

CERTIFICATIONS

- Machine Learning Specialization; Deep Learning Specialization; TensorFlow Developer
- AWS Certified Machine Learning; Google Professional Machine Learning Engineer; Microsoft Azure AI Engineer
- Kubernetes Application Developer; Databricks Machine Learning; NVIDIA Deep Learning Institute

ADDITIONAL EXPERTISE

End-to-End ML System Design • Production ML Architecture • LLM Application Architecture • AI Agent Architecture • Model Serving & Inference Optimization • GPU Computing • Distributed Training • Mixed Precision Training • CUDA Fundamentals • Responsible AI • AI Safety • Explainability • Fairness & Bias Detection • Privacy-Preserving ML • Scalability • Cost Optimization • Performance Engineering